Tutorial 3: Solutions

ES 336 – Computer Organization and Architecture

Indian Institute of Technology Gandhinagar

Problem 1

Two processors, P1 and P2, execute the same program.

- Processor P1: Clock rate = 2.5 GHz, Average CPI = 2.0
- Processor P2: Clock rate = 3.0 GHz, Average CPI = 3.0

Solution 1

(a) Compute the MIPS rating of each processor.
The formula for MIPS is:

$$\text{MIPS} = \frac{\text{Clock Rate}}{\text{CPI} \times 10^6}$$

For Processor P1:

$$\text{MIPS}_{P1} = \frac{2.5 \times 10^9}{2.0 \times 10^6} = \frac{2500}{2.0} = 1250 \text{ MIPS}$$

For Processor P2:

$$\text{MIPS}_{P2} = \frac{3.0 \times 10^9}{3.0 \times 10^6} = \frac{3000}{3.0} = 1000 \text{ MIPS}$$

(b) Which processor has the higher MIPS rating?
Processor **P1** has the higher MIPS rating (1250 vs 1000).
(c) Which processor executes the program faster?
Let I be the number of instructions in the program.

$$\text{Execution Time} = \frac{\text{Instruction Count} \times \text{CPI}}{\text{Clock Rate}}$$

Time for P1 (T_{P1}):

$$T_{P1} = \frac{I \times 2.0}{2.5 \times 10^9} = I \times 0.8 \times 10^{-9} \text{ seconds}$$

Time for P2 (T_{P2}):

$$T_{P2} = \frac{I \times 3.0}{3.0 \times 10^9} = I \times 1.0 \times 10^{-9} \text{ seconds}$$

Since $0.8 < 1.0$, **Processor P1** is faster.

Problem 2

Consider two different implementations, P1 and P2, of the same instruction set architecture.

- Processor P1: Clock Rate = 2.5 GHz
- Processor P2: Clock Rate = 3.0 GHz

Class	Freq	CPI (P1)	CPI (P2)
A	10%	1	2
B	20%	2	2
C	50%	3	2
D	20%	3	2

Solution 2

1. Calculate the global Average CPI for each processor.

$$\text{Average CPI} = \sum(\text{Frequency}_i \times \text{CPI}_i)$$

For P1:

$$\text{CPI}_{P1} = (0.1 \times 1) + (0.2 \times 2) + (0.5 \times 3) + (0.2 \times 3)$$

$$\text{CPI}_{P1} = 0.1 + 0.4 + 1.5 + 0.6 = \mathbf{2.6}$$

For P2:

$$\text{CPI}_{P2} = (0.1 \times 2) + (0.2 \times 2) + (0.5 \times 2) + (0.2 \times 2)$$

$$\text{CPI}_{P2} = 0.2 + 0.4 + 1.0 + 0.4 = \mathbf{2.0}$$

2. Calculate execution time ($IC = 1.0 \times 10^6$ instructions).

$$\text{Time} = \frac{\text{IC} \times \text{CPI}}{\text{Clock Rate}}$$

For P1:

$$T_{P1} = \frac{10^6 \times 2.6}{2.5 \times 10^9} = \frac{2.6}{2500} = 0.00104 \text{ s} = \mathbf{1.04 \text{ ms}}$$

For P2:

$$T_{P2} = \frac{10^6 \times 2.0}{3.0 \times 10^9} = \frac{2.0}{3000} \approx 0.00067 \text{ s} = \mathbf{0.67 \text{ ms}}$$

3. Which processor is faster? Calculate Speedup.

P2 is faster (lower execution time).

$$\text{Speedup} = \frac{T_{P1}}{T_{P2}} = \frac{1.04}{0.667} \approx \mathbf{1.56}$$

Processor P2 is 1.56 times faster than P1.

Problem 3

Four benchmark programs are executed on three computers. Execution time in seconds (for 10^8 instructions):

Program	Comp A (s)	Comp B (s)	Comp C (s)
Prog 1	1	10	20
Prog 2	1000	100	20
Prog 3	500	1000	50
Prog 4	100	800	100

Solution 3

1. Calculate MIPS ($IC = 100 \times 10^6$).

$$\text{MIPS} = \frac{100 \times 10^6}{\text{Time} \times 10^6} = \frac{100}{\text{Time}}$$

- **Computer A:** P1: $100/1 = 100$, P2: $100/1000 = 0.1$, P3: $100/500 = 0.2$, P4: $100/100 = 1$.
- **Computer B:** P1: $100/10 = 10$, P2: $100/100 = 1$, P3: $100/1000 = 0.1$, P4: $100/800 = 0.125$.
- **Computer C:** P1: $100/20 = 5$, P2: $100/20 = 5$, P3: $100/50 = 2$, P4: $100/100 = 1$.

2. Calculate Arithmetic and Harmonic Means of MIPS.

Computer A:

$$AM_A = \frac{100 + 0.1 + 0.2 + 1}{4} = 25.325$$

$$HM_A = \frac{4}{\frac{1}{100} + \frac{1}{0.1} + \frac{1}{0.2} + \frac{1}{1}} = \frac{4}{0.01 + 10 + 5 + 1} = \frac{4}{16.01} \approx 0.25$$

Computer B:

$$AM_B = \frac{10 + 1 + 0.1 + 0.125}{4} = 2.806$$

$$HM_B = \frac{4}{\frac{1}{10} + \frac{1}{1} + \frac{1}{0.1} + \frac{1}{0.125}} = \frac{4}{0.1 + 1 + 10 + 8} = \frac{4}{19.1} \approx 0.21$$

Computer C:

$$AM_C = \frac{5+5+2+1}{4} = 3.25$$

$$HM_C = \frac{4}{\frac{1}{5}+\frac{1}{5}+\frac{1}{2}+\frac{1}{1}} = \frac{4}{0.2+0.2+0.5+1} = \frac{4}{1.9} \approx 2.10$$

3. Rank the computers.

- **Arithmetic Mean:** $A > C > B$
- **Harmonic Mean:** $C > A > B$

Problem 4

Comparison of System X against a Reference Machine.

Benchmark	Reference (s)	System X (s)
Prog A	1000	500
Prog B	2500	2000
Prog C	500	250

Solution 4

1. Calculate Speedup Ratio (Ref / Sys X).

- Prog A: $1000/500 = 2.0$
- Prog B: $2500/2000 = 1.25$
- Prog C: $500/250 = 2.0$

2. Arithmetic Mean of execution times.

- Ref Mean: $(1000 + 2500 + 500)/3 = 1333.33$ s
- Sys X Mean: $(500 + 2000 + 250)/3 = 916.67$ s

System X is faster by factor: $1333.33/916.67 \approx 1.45$.

3. Geometric Mean of Speedup Ratios.

$$GM = \sqrt[3]{2.0 \times 1.25 \times 2.0} = \sqrt[3]{5.0} \approx 1.71$$

4. Comparison. The Geometric Mean (1.71) is generally preferred (like in SPEC) because it provides a consistent measure regardless of the reference machine used or total running time normalization. The arithmetic mean of execution times (1.45) is biased by the longest-running program (Prog B), where the speedup was lowest.

Problem 5

Given:

- Computation enhancement speedup (S_e) = 10
- Fraction of time in computation (f_e) = 40% = 0.4
- Fraction of time waiting for I/O = 60% = 0.6

Solution 5

Using Amdahl's Law:

$$S_{\text{overall}} = \frac{1}{(1 - f_e) + \frac{f_e}{S_e}}$$

$$S_{\text{overall}} = \frac{1}{0.6 + \frac{0.4}{10}} = \frac{1}{0.6 + 0.04} = \frac{1}{0.64} = 1.5625$$

Answer: The overall speedup is **1.56**.

Problem 6

Given:

- FP operations: 40%, Enhanced by 5×
- Memory accesses: 30%, Enhanced by 2×
- Control/Other: 30%, Unchanged

Solution 6

(a) Both enhancements applied:

$$S_{both} = \frac{1}{0.3 + \frac{0.3}{2} + \frac{0.4}{5}} = \frac{1}{0.3 + 0.15 + 0.08} = \frac{1}{0.53} \approx 1.89$$

(b) Only FP enhancement applied:

$$S_{FP} = \frac{1}{(0.3 + 0.3) + \frac{0.4}{5}} = \frac{1}{0.6 + 0.08} = \frac{1}{0.68} \approx 1.47$$

(c) Compare with only Memory enhancement:

$$S_{Mem} = \frac{1}{(0.3 + 0.4) + \frac{0.3}{2}} = \frac{1}{0.7 + 0.15} = \frac{1}{0.85} \approx 1.18$$

Answer: The FP enhancement provides greater benefit (1.47 vs 1.18) because the program spends more time in FP operations (40%) than memory (30%) and the speedup factor for FP is higher.

Problem 7

Given:

- Serial fraction $(s) = 0.10$.
- Parallel fraction $(p) = 0.90$.
- Number of processors $(N) = 32$.

Solution 7

The total execution time is normalized to 1.
Thus:

$$\text{Serial fraction} = s = 0.10, \qquad \text{Parallel fraction} = p = 0.90$$

(a) Amdahl's Law (Fixed Problem Size)

Amdahl's Law assumes a fixed problem size and is given by:

$$S_A = \frac{1}{(1-p) + \frac{p}{N}}$$

Substituting the given values:

$$S_A = \frac{1}{0.10 + \frac{0.90}{32}} = \frac{1}{0.128125} \approx \boxed{7.8}$$

The maximum possible speedup as $N \to \infty$ is:

$$S_{A,\max} = \frac{1}{s} = \boxed{10}$$

(b) Gustafson's Law (Scaled Problem Size)

Gustafson's Law assumes fixed execution time and is given by:

$$S_G = (1-p) + pN$$

Substituting the given values:

$$S_G = 0.10 + (0.90 \times 32) = \boxed{28.9}$$

(c) Explanation

Amdahl's Law uses the fraction of serial execution time measured on a single processor and assumes a fixed problem size, which limits speedup. Gustafson's Law assumes a fixed execution time and allows the parallel workload to scale with the number of processors, leading to much higher speedup.

Problem 8

Given (Single-Core Baseline):

- Average CPI = 2.24
- MIPS = 178

The instruction mix is:

Instruction Type	Frequency	CPI
Arithmetic / Logic	0.60	1
Load / Store (Cache Hit)	0.18	2
Branch	0.12	4
Memory Reference (Cache Miss)	0.10	8

Solution 8

a. Determine the average CPI for the parallel tasks. The instruction mix is the same, but the Memory Reference (Cache Miss) CPI changes.

$$\text{New CPI} = (0.6 \times 1) + (0.18 \times 2) + (0.12 \times 4) + (0.10 \times 12)$$

$$\text{New CPI} = 0.6 + 0.36 + 0.48 + 1.2 = \mathbf{2.64}$$

(b) New MIPS Rate per Core

Since clock rate remains unchanged, MIPS is inversely proportional to CPI:

$$\text{MIPS}_{\text{new}} = \text{MIPS}_{\text{old}} \times \frac{\text{CPI}_{\text{old}}}{\text{CPI}_{\text{new}}}$$

$$\text{MIPS}_{\text{new}} = 178 \times \frac{2.24}{2.64} \approx \boxed{151}$$

(c) Speedup Calculation

Let the total instruction count be $I = 10^8$.

The execution time on the original single-core processor is:

$$T_{\text{old}} = \frac{I \times \text{CPI}_{\text{old}}}{F} = \frac{10^8 \times 2.24}{F}$$

In the parallel system, the program is divided equally among 8 cores. Hence, each core executes $\frac{I}{8}$ instructions. In addition, each task incurs a serial overhead of 25,000 instructions.

$$\frac{I}{8} = \frac{10^8}{8} = 12.5 \times 10^6$$

Thus, the execution time on the parallel system is:

$$T_{\text{new}} = \frac{(12.5 \times 10^6 + 25{,}000) \times 2.64}{F}$$

The overall speedup is given by:

$$\text{Speedup} = \frac{T_{\text{old}}}{T_{\text{new}}} = \frac{10^8 \times 2.24}{(12.5 \times 10^6 + 25{,}000) \times 2.64} \approx 6.75$$

$$\boxed{\text{Overall Speedup} \approx 6.75}$$

(d) Amdahl's Law and Gustafson's Law

(i) Serial Fraction Total serial (synchronization) instructions:

$$8 \times 25{,}000 = 200{,}000$$

Total instructions executed:

$$10^8 + 200{,}000 = 100{,}200{,}000$$

Thus, the serial fraction is:

$$s = \frac{200{,}000}{100{,}200{,}000} \approx \boxed{0.002}$$

Parallel fraction:

$$p = 1 - s \approx 0.998$$

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(ii) Speedup using Amdahl's Law

$$S_A = \frac{1}{s + \frac{p}{N}} = \frac{1}{0.002 + \frac{0.998}{8}} \approx \boxed{7.85}$$

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(iii) Speedup using Gustafson's Law

$$S_G = (1 - p) + pN = s + (1 - s)N = 0.002 + (0.998 \times 8) \approx \boxed{7.986}$$

Under scaled workload interpretation, this corresponds to an effective speedup close to:

$$\boxed{8}$$

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(iv) Explanation The observed speedup is lower than the ideal value due to increased CPI caused by memory contention and additional synchronization overhead. Amdahl's Law assumes a fixed problem size and highlights the limiting effect of even small serial fractions. Gustafson's Law assumes fixed execution time and allows the parallel workload to scale with the number of processors, resulting in near-linear speedup.

Homework Problems

Problem A

Given:

- Enhanced mode speedup (S_e) = 10.
- Enhanced mode is used 50% of the time, measured as a fraction of execution time on the enhanced system.

Solution

Let T_{new} be the total execution time on the enhanced computer.
Since the enhanced mode is used for 50% of the execution time:

$$T_{\text{enhanced,new}} = 0.5T_{\text{new}}, \qquad T_{\text{unenhanced,new}} = 0.5T_{\text{new}}$$

The enhanced mode provides a speedup of 10, hence:

$$T_{\text{enhanced,new}} = \frac{T_{\text{enhanced,old}}}{10} \Rightarrow T_{\text{enhanced,old}} = 10 \times 0.5T_{\text{new}} = 5T_{\text{new}}$$

The unenhanced portion is unaffected:

$$T_{\text{unenhanced,old}} = T_{\text{unenhanced,new}} = 0.5T_{\text{new}}$$

Therefore, the original execution time is:

$$T_{\text{old}} = 5T_{\text{new}} + 0.5T_{\text{new}} = 5.5T_{\text{new}}$$

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(a) Speedup obtained from fast mode

$$\text{Speedup} = \frac{T_{\text{old}}}{T_{\text{new}}} = \frac{5.5T_{\text{new}}}{T_{\text{new}}} = \boxed{5.5}$$

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(b) Percentage of original execution time converted to fast mode

$$f = \frac{T_{\text{enhanced,old}}}{T_{\text{old}}} = \frac{5T_{\text{new}}}{5.5T_{\text{new}}} = \frac{5}{5.5} \approx \boxed{90.9\%}$$

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(c) Comparison using Amdahl's Law and a scaled-time interpretation

Amdahl's Law uses the fraction of the *original* execution time that can be enhanced. With $f \approx 0.909$ and $S_e = 10$:

$$S_A = \frac{1}{(1-f) + \frac{f}{S_e}} = \frac{1}{0.091 + 0.0909} \approx \boxed{5.5}$$

A **scaled-time interpretation** may also be applied by using the fraction of execution time measured on the enhanced system ($f' = 0.5$):

$$S = (1 - f') + f' \times S_e = 0.5 + (0.5 \times 10) = \boxed{5.5}$$

Both approaches yield the same speedup because the fractions of execution time are derived consistently. The latter expression should be viewed as a consistency check based on the enhanced execution profile, rather than a full parallel scaling application of Gustafson's Law.The scaling application of Gustafson's was discussed in Problem 7.